Building the ML model — pika SDM

Data & target

- Binary classification task: presence (1) vs absence (0) of pika, predicted from a site's coordinates.
- From 18,184 rows down to 17,187 after dropping incomplete rows; a ≥1970 temporal filter applied to *both* classes for consistency with the climate baseline → ~13,700 presences, 220 absences.
- Absences enriched by integrating 82 extirpated Great Basin sites (Millar 2018) + 10 USGS sites, after duplicate checks (haversine distance) and coordinate validation.

Features (11)

- elevation, slope_deg, bio1, bio4, bio5, bio6, bio12, bio15, months_with_snow, winter_precip, estim_snow.
- All derived from coordinates (WorldClim 1970-2000 baseline for climate; snow proxies from monthly tmax/precip).
- Key choice: lat/long are NOT used as features, so the model learns environmental conditions rather than geography. This avoids a geographic shortcut and keeps the pipeline projectable to future climate.

Model

- Downsampling ensemble: 100 Random Forests, each trained on all training absences + an equal-sized random sample of presences (a balanced set).
- Final prediction = mean of the 100 models' probabilities.
- Rationale: handles the ~60:1 class imbalance directly, and averaging many models reduces variance.

Validation & metrics

- Stratified 80/20 split. ROC-AUC = 0.997, but this is inflated by imbalance and spatial autocorrelation, so it's not the metric we rely on.
- PR-AUC on the absence class = 0.881 (random baseline 0.015), i.e. ~60× better than chance — a far more informative metric for the rare class.
- Spatial validation (hold out whole geographic blocks): PR-AUC drops to mean 0.55, min 0.15. The gap between 0.88 and 0.55 is the real finding — the model generalises poorly to unseen regions, meaning climate alone doesn't explain where pikas disappear.
- Decision threshold tuned deliberately (not left at 0.5): trading off recall on absences vs false alarms, depending on whether the goal is an alarm system (recall 1.00) or a map (higher precision).

Future projection

- Same sites, same model: each site's climate features are replaced with values projected for ~2050 (SSP2-4.5), location fixed, then suitability is re-predicted.
- Reported as a continuous suitability gradient (not a binary extinct/present call): mean suitability 0.95 → 0.83, ~90% of sites decline, ~22% decline sharply. Loss concentrates on warm southern/low-elevation margins.

Bottom line

- The model reliably estimates an aggregate gradient of climatic suitability, but not the fate of any single site — because of the imbalance, weak spatial generalisation, and the fact that climate isn't the only driver of extirpations.

Data sources

- WorldClim v2.1 — historical climate 1970-2000 (bioclimatic variables): https://www.worldclim.org/data/worldclim21.html
- WorldClim — future climate projections, CMIP6 / SSP scenarios incl. SSP2-4.5: https://www.worldclim.org/data/cmip6/cmip6climate.html